The Must-Learn 21

Exam-ready answers for the questions most likely to come in **CIE-1**. Built from **all 179 class slides** and the **Nov 2024 CIE-1 paper**. Written in **4 · 5 · 6-mark** shape so you can reproduce them in the hall. Every list comes with a **memory line**: the first letter of each word is the answer.

Unit I · Intro · Lit review · Citations

Unit II · Research & experimental design

Unit III · Data collection · Sampling · Scaling (slides 1–54)

Not in CIE-1: Data Analysis (U3 slides 55–80) · Units IV–V

21	13	10	25+
core questions 7 per unit, full answers	rapid-fire 3–5 markers compact answers	L3 scenario drills the "apply" questions	target out of 30 2 questions × 15

★★★ Banker: asked before or a slide example. Learn it word for word.

★★ Very likely. Core slide content not yet asked.

★ Good to know. Short-answer insurance.

Highlighted words are the keywords the examiner scans for. Underline them in your answer.

Start with page 2 (how the paper works), then page 3 (how to study). Do the ★★★ questions first.

Decoding the CIE-1 paper

Nov 2024 CIE-1: 60 minutes, 30 marks, **answer any two full questions** out of three, each worth 15. Portions were lectures L1–L14 (Units I and II only). Here is where every part came from.

Q	Question (2024)	Unit	Marks	Level	Source
1a	Briefly explain the different types of research	U1	6	L2	SLIDES U1 s.19–21
1b	Define independent, dependent, extraneous, continuous & discrete variables	U2	5	L1	SLIDES + KOTHARI U2 s.6–9
1c	Key differences between engineering research and research in other fields	U1	4	L3	TEXTBOOK Deb et al.
2a	Why is ethics important in research? Key ethical considerations	U1	6	L2	SLIDES U1 s.22–24
2b	3×2×2 factorial (therapy × duration × gender): list all combinations	U2	6	L3	SLIDE, WORD FOR WORD U2 s.48
2c	Your team must frame a project title: what will you consider?	U1	3	L3	SLIDES U1 s.36–38
3a	List and explain three main functions of citation	U1	6	L2	SLIDE U1 s.34
3b	Explain RBD; design it for F1, F2, F3 on rice, wheat, jowar	U2	6	L3	SLIDE EXAMPLE, TWEAKED U2 s.33
3c	Compare critical and creative reading	U1	3	L2	SLIDE U1 s.32

1 · The slides are the syllabus

Seven of the nine parts came straight off the class slides. 1b was mostly slides (only "continuous / discrete" is from Kothari). Only 1c needed the textbook. **If it's on a slide, it can be asked.**

2 · Slide examples get recycled

The 3×2×2 therapy question is **copied word for word** from Unit 2 slide 48. The RBD question reuses slide 33's fertiliser example with the crops swapped (rice, corn, wheat became rice, wheat, jowar). **Learn every worked example on the slides, then expect a twist.**

3 · Every question mixes units

All three questions were built as **U1 + U2 + U1**. You can't dodge a unit by choosing questions: a skipped unit costs you marks in *every* question.

4 · 42% of the marks were "apply" (L3)

1c, 2b, 2c and 3b were L3: 19 of the 45 marks on the paper. They give you a scenario and ask you to design, list or analyse. That's why this pack has **scenario drills** (page 29).

This year's prediction

Portions grew: **Unit III up to slide 54** is now in. So expect each 15-mark question to carry one part from each unit (for example U1 6 + U2 5 + U3 4, rotated). Unit I drops from six parts to about three, and Unit III gets about three. Last year's 6-mark Unit II questions were factorial and RBD. **Latin Square is the one design not yet asked**, which makes it the strongest single candidate. Repeats of RBD and factorial are still very possible: teachers reuse papers.

Marks split. 6 + 5 + 4 or 6 + 6 + 3. A 6-marker is 40% of a question: that's where answers win or lose.

The 25+ maths. You write 30 marks. To score 25 you can lose at most 5, so every part needs complete lists, an example and a diagram or table.

An honest note. Nobody can promise 100%. But every 2024 question came from material in this pack, and it covers every CIE-1 slide.

How to study this (and how to write it)

If you have 3 days

1. **Day 1: Unit II.** It's the most scoring (definitions plus designs you can draw). Draw the RBD, LSD and factorial layouts from memory twice.
2. **Day 2: Unit III**, then the Unit I ★★★ questions. Make the comparison tables your own.
3. **Day 3:** rest of Unit I, the rapid-fire answers and the scenario drills. Then a **timed mock**: one full 15-mark question in 25 minutes.

Night before: memory lines (page 30) and the cheat sheet (page 33), then sleep.

If you have one night (about 6 hours)

1. **Hours 1–2:** Unit II ★★★: variables, hypothesis, RBD, LSD, factorial.
2. **Hour 3:** Unit I ★★★: types of research, ethics, citation functions, critical vs creative reading, title.
3. **Hour 4:** Unit III ★★★: primary vs secondary, questionnaire vs schedule, sampling definitions, sample design types, NOIR scales.
4. **Hour 5:** all the ★★ questions, skeleton only (headings plus keywords).
5. **Hour 6:** memory lines, cheat sheet, drills, then sleep.

The method: active recall, not re-reading

1 · Read once

Read the answer slowly. Say the example out loud in your own words.

2 · Close and write

Cover the page and write the **skeleton**: definition, headings, keywords. Use the **memory line** to pull each list back.

3 · Check and repeat

Mark what you missed and redo it. Come back the next day: the second recall is what makes it stick.

The answer formula: write to the marks

Marks	Structure that gets full marks	Length (handwritten)
6 M	1-line **definition**, then 5–6 **headed points**, then an **example**, then a **diagram or table**, then a 1-line wrap-up	1 to 1½ pages
5 M	Definition, then 4–5 headed points, then an example	About 1 page
4 M	4 crisp points, **or** a 4-row comparison table	About ¾ page
3 M	3 points or a 3-row table. No essays.	About ½ page

In the exam hall

- **First 5 minutes:** read all three questions. Pick the two where your **6-markers** are strongest.
- **Clock:** 5 min choosing, then 25 min for question A, 25 min for question B, then 5 min checking.
- Never leave a part blank. A correct definition alone earns 1–2 marks.
- Number every part exactly as in the paper (2a, 2b...).

Presentation rules

- **"Differentiate" or "compare" means a table**, every time.
- **A design question means a diagram:** RBD and LSD grids, the factorial table, the CRD flow.
- **Definitions in slide wording.** The examiner marks against the slides.
- **Underline** the keywords and give every point a heading.
- **L3 questions:** restate the scenario, name the concept, apply it (layout, list or table).

UNIT I · CO1

Research Methodology

Meaning, objectives and types of research · Ethics and misconduct · Literature review and technical reading · Citations, title and keywords · Acknowledgments

47	6 of 9	~3 parts
slides	parts in 2024	expected this year

Q1 UNIT I — Briefly explain the different types of research.

6 M · ★★★ Banker · PYQ 2024 · Q1a · Slides 19–21

Research is classified in pairs, depending on its purpose, the kind of data it uses, and where it is carried out.

1. Descriptive vs Analytical. Descriptive research includes surveys and fact-finding enquiries; it describes the state of affairs as it exists, and the researcher has no control over the variables (*e.g.* a survey of how many students use UPI daily). In analytical research the researcher uses facts or information already available and analyses them to make a critical evaluation (*e.g.* analysing published accident data to find the main causes).

2. Applied vs Fundamental. Applied (action) research aims at finding a solution for an immediate problem facing society or an industrial or business organisation (*e.g.* reducing the rejection rate on a production line). Fundamental (basic or pure) research is concerned with generalisations and the formulation of a theory (*e.g.* studying the properties of graphene).

3. Quantitative vs Qualitative. Quantitative research is based on the measurement of quantity or amount (*e.g.* battery life in hours). Qualitative research is concerned with phenomena relating to quality or kind, such as motives, opinions and attitudes (*e.g.* why customers prefer a brand).

4. Conceptual vs Empirical. Conceptual research relates to some abstract idea or theory; thinkers use it to develop new concepts or reinterpret existing ones. Empirical research is data-based: its conclusions can be verified by observation or experiment, so it is also called *experimental* research.

5. Some other types.

- **One-time vs longitudinal:** confined to a single time period vs carried on over several time periods.
- **Field-setting, laboratory or simulation:** classified by the environment the research is done in.
- **Clinical or diagnostic:** uses case-study or in-depth methods to reach basic causal relations.
- **Conclusion-oriented vs decision-oriented:** in the first, the researcher is free to pick a problem and redesign the enquiry as they go; the second serves the needs of a decision-maker.

MEMORY LINE **Dad Always Asks For Quick Questions, Clear Explanations**
Descriptive · Analytical · Applied · Fundamental · Quantitative · Qualitative · Conceptual · Empirical

MEMORY LINE **Only Lazy Folks Love Sleeping, Coffee Cures Drowsiness**
One-time · Longitudinal · Field-setting · Laboratory · Simulation · Clinical / diagnostic · Conclusion-oriented · Decision-oriented

Basis	Type A	Type B
Purpose of study	Descriptive	Analytical
Use of results	Applied	Fundamental
Nature of data	Quantitative	Qualitative
Approach	Conceptual	Empirical

Draw this summary table at the end of your answer. It's a quick mark and shows the classification at a glance.

Examiner's eye

Use the 4 pairs as headings, each with a **one-line example**. That alone gets 5/6. The "other types" line is your sixth mark.

Q2
UNIT I

Why is ethics important in research? What key ethical considerations should researchers keep in mind?

6 M · ★★★ Banker · PYQ 2024 · Q2a · Slides 22-24

Definition. Ethics in research provides guidelines for the responsible conduct of research: what researchers should and should not do while planning, doing and reporting research.

Why ethics is important

- **Protects the integrity of research.** It prevents fabrication, falsification and plagiarism, so the research record stays true.
- **Builds trust and credibility.** The public, funders and other researchers rely on published results.
- **Prevents harm.** Misconduct can cause harm to people and the environment, wastes resources and damages the credibility of research.
- **Supports collaboration.** Fairness, mutual respect and honest credit-sharing let teams work together.
- **Ensures accountability.** Research uses public money and must follow laws and institutional policies.

MEMORY LINE **I Trust Parents' Common Advice**
Integrity of research · Trust and credibility · Prevents harm · Collaboration · Accountability

Key ethical principles (the slide's list)

Honesty	Report data, results and methods honestly. Never fabricate, falsify or misrepresent.
Objectivity	Avoid bias in design, analysis, interpretation and peer review. Disclose conflicts of interest.
Integrity	Keep promises and agreements. Act with sincerity and consistency.
Carefulness	Avoid careless errors. Keep good records of all research activities.
Openness	Share data, results, ideas and tools. Be open to criticism.
Respect for IP	Honour patents and copyrights. Give proper credit (cite). Never plagiarise.
Confidentiality	Protect confidential material: papers under review, participant data, personnel records.

MEMORY LINE **Honest Officers In Court Openly Respect Confidentiality**
Honesty · Objectivity · Integrity · Carefulness · Openness · Respect for IP · Confidentiality

Also mention: **responsible publication**, **social responsibility** and **protection of human and animal subjects**.

Examiner's eye
The question has two halves ("why" and "considerations"). Give each a heading or you lose marks. Close with one line: *breaking these principles is research misconduct (FFP)*.

Q3
UNIT I

What is research misconduct? Explain its types and the common forms of plagiarism.

5 M · ★★ Very likely · Slides 23-25

Definition. Research misconduct is actions or questionable research practices that fall short of the standards of ethics, research and scholarship required to uphold the integrity of research. It harms people and the environment, wastes resources, undermines the research record and damages credibility.

Types of misconduct: FFP

Fabrication: making up data or results and recording or reporting them (*e.g.* inventing readings for a test never performed).

Falsification: manipulating research materials, or changing or omitting data or results, so the research is not accurately represented (*e.g.* deleting points that don't fit the hypothesis).

Plagiarism: the appropriation of another person's ideas, processes, results or words without giving appropriate credit.

MEMORY LINE **Fake Facts Punished**
Fabrication · **F**alsification · **P**lagiarism

Research misconduct does not include honest error or differences of opinion.

Forms of plagiarism

Direct	Taking another person's work and passing it off as your own without changing anything.
Self-plagiarism	Reusing your own earlier published work without acknowledging it.
Mosaic	Partly paraphrasing, so the paper is a mix of the sources' words and your own.
Paraphrasing without credit	Changing the wording of a source but not citing it.
Accidental	Using information from sources but forgetting to include citations.
Misattribution	Attributing information from one source to a different source.

MEMORY LINE **Don't Steal My Project And Marks**
Direct · **S**elf-plagiarism · **M**osaic · **P**araphrasing without credit · **A**ccidental · **M**isattribution

Q4 UNIT I

List and explain the three main functions of citation.

6 M | ★★★ Banker | PYQ 2024 · Q3a | Slides 33-35, 41-42

Citations (references) credit others for their work while allowing readers to trace the source publication. Each source is cited twice: (1) an **in-text citation** exactly where it is quoted or paraphrased, and (2) an entry in the **reference list** at the end of the article or chapter.

1. Verification function. Authors may make intentional or unintentional distortions or misleading statements. A citation lets readers check whether the original source justifies the claim, and whether the claim is properly described in the present work.

2. Acknowledgment function. Researchers primarily receive credit for their work through citations. Citations play a crucial role in promotion and continued employment, and organisations award research funding based on reputation. They enhance a researcher's reputation and give the background of the work.

3. Documentation function. Citations document scientific concepts and the historical progress of a technology over the years.

MEMORY LINE **Very Accurate Data**
Verification · **A**cknowledgment · **D**ocumentation

When citations fail their purpose (bonus point)

- **Spurious:** citing work you haven't read, or that isn't relevant.
- **Biased:** citing only work that supports your view, or your friends' work.
- **Self-citations:** excessive citing of your own work to inflate counts.
- **Coercive:** an editor or reviewer pressuring authors to cite their journal or work.

MEMORY LINE **Some Bad Scholars Cheat**
Spurious · **B**iased · **S**elf-citations · **C**oercive

EXAMPLE · IEEE STYLE
In text: "...as shown in **[1]**." → Reference list: **[1]** D. Deb, R. Dey, and V. E. Balas, *Engineering Research Methodology*. Singapore: Springer, 2019.
Other styles on the slides: **ASCE** (civil) and **ASME** (mechanical).

Examiner's eye
The question says "list and explain", so **name all three first** in one line, then explain each. The four failed-citation types are the extra that turns 5/6 into 6/6.

Q5 UNIT I

What should a literature review achieve? Explain new and existing knowledge and the analysis and synthesis of prior art.

6 M | ★★ Very likely | Slides 26-31

New and existing knowledge. New knowledge can only be interpreted within the context of what is already known; it cannot exist without that foundation. To show your work is original, you describe what already exists and point out the missing part. Existing knowledge gives you the context, the significance, the originality and the tools.

MEMORY LINE **Cats Sleep On Tables**
Context · **S**ignificance · **O**riginality · **T**ools

What a literature review must do.

- Summarise what is already known (the state of the art).
- Detail the key concepts, main factors or parameters and the relationships between them.
- Describe complementary existing approaches.
- List the inconsistencies or shortcomings in published work.
- Identify results that are inconclusive or contradictory.
- Give a compelling reason to do further work (the research gap).

MEMORY LINE **Smart Kids Can Identify Interesting Research**

Summarise the state of the art · **K**ey concepts and factors · **C**omplementary approaches · **I**nconsistencies and shortcomings · **I**nconclusive or contradictory results · **R**eason for further work

The two-step process. (i) Identify the major topics, subtopics or concepts relevant to the subject. (ii) Place the citation of each source (article, patent, website, data) in the right category, for example with a √. This gives a **concept matrix**:

Source	Sensor design	Low power	Field testing	Cost analysis
Paper [1]	√	√		
Paper [2]	√		√	
Patent [3]	√	√		

An empty column, like "Cost analysis" here, is the research gap.

Analysis and synthesis of prior art. The researcher should work through these steps:

1. Understanding the hypothesis
2. Understanding the models and experimental conditions used
3. Making connections
4. Comparing and contrasting the information
5. Finding the strong points and the loopholes

MEMORY LINE **Hungry Monkeys Can Crack Shells**

Hypothesis · **M**odels and experimental conditions · **C**onnections · **C**ompare and contrast · **S**trong points and loopholes

Analysis breaks each work down; *synthesis* combines the works into one picture that reveals the gap.

Supporting tools. **Bibliographic databases** are abstracting and indexing services that collect citation information and abstracts and make them searchable (Web of Science, Google Scholar). **Conceptualising research** means having a significant problem, the knowledge that will address it, and a possible way to make that new knowledge.

MEMORY LINE **Smart Kids Win**

Significant problem · **K**nowledge that will address it · **W**ay to make that new knowledge

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Q6 — Compare critical reading and creative reading.

UNIT I

4 M · ★★★ Banker · PYQ 2024 · Q3c · Slide 32

Reading a research paper is a critical process, but a good researcher reads it in two ways.

Basis	Critical reading	Creative reading
Aim	To find mistakes, weaknesses and loopholes	To find good ideas: other applications, generalisations, extended work
Attitude	Sceptical and evaluative	Constructive and exploratory
Questions asked	Are the assumptions valid? Is the evidence enough? Is the method sound?	What did the authors miss? Are there plausible modifications that raise new practical challenges?
Difficulty	Relatively easy	Harder: you must actively look for what isn't written
Output	A critique: list of flaws and limitations	New research ideas and problems

MEMORY LINE **Always Ask Questions, Don't Overthink**

Aim · **A**ttitude · **Q**uestions asked · **D**ifficulty · **O**utput

Conclusion: critical reading stops you building on wrong results; creative reading gives you your research problem. Use both.

Q7
UNIT I

Your team must frame a title for its research project. What will you consider? (Also asked as: "Explain the impact of title and keywords on citations.")

3-4 M · ★★★ Banker · PYQ 2024 · Q2c · Slides 36-38

The citation rate of a paper depends on the significance and availability of the journal, the publication type, the research area and the importance of the work. It also depends on the length of the title, the type of title and the keywords chosen. The title is the most important attribute of a paper: it markets the paper and makes it traceable.

1. **Informative and accurate.** It should represent the paper effectively and match its scope: *problem + method (+ application)*.
2. **Type of title.** *Descriptive* (states the topic), *declarative* (states the main finding) or *interrogative* (asks a question). Choose deliberately.
3. **Length.** Concise. Cut filler like "A study on...", but don't make it so short that it's vague.
4. **Specific markers.** A colon or hyphen to add a subtitle; a question mark for an interrogative title.
5. **Keywords.** Use the words people actually search. Search engines, journals, digital libraries and indexing services use keywords to categorise the work and direct it to the relevant audience.
6. **Gains attention, without hype.** Catchy but not misleading. Avoid jargon and unexplained abbreviations.

MEMORY LINE **I Totally Love Mom's Kheer, Guys**

Informative and accurate · **T**ype of title · **L**ength · **M**arkers · **K**eywords · **G**ains attention

MEMORY LINE **Tea Lovers Meet**

Type of title · **L**ength of title · **M**arkers (colon, hyphen, ?)

EXAMPLE TITLE FOR YOUR PROJECT

"Low-Cost IoT Soil-Moisture Monitoring for Small Farms: Design and Field Evaluation": problem, method and application, keyword-rich, with a colon subtitle.

Examiner's eye

For 3 marks: the three aspects from the slide (**type, length, specific markers**) plus one line on **keywords**, and write an example title. Examples make L3 answers score.

Rapid-fire • Unit I

Compact answers for 3–5 mark parts

Meaning and objectives of research

4-5 M ★★ Slides 13-18

Re + search: "again" + "to examine closely". Research is a systematised effort to gain new knowledge, an art of scientific investigation, a movement from the known to the unknown.

Research comprises: defining the problem → formulating a hypothesis → collecting, organising and evaluating data → making deductions → testing the conclusions.

Do Have Coffee During Tests

Define the problem · Hypothesis · Collect, organise and evaluate data · Deductions · Test the conclusions

Four objectives:

1. Gain familiarity or new insights (exploratory)
2. Portray characteristics accurately (descriptive)
3. Determine frequency or association (diagnostic)
4. Test a causal hypothesis (hypothesis-testing)

Every Day Dreams Happen

Exploratory (familiarity, new insights) · Descriptive (portray characteristics) · Diagnostic (frequency, association) · Hypothesis-testing (causal)

Motivations: a research degree, the challenge of unsolved problems, intellectual joy, service to society, respectability.

Degree Can Inspire Some Respect

Degree and its benefits · Challenge of unsolved problems · Intellectual joy · Service to society · Respectability

Engineering research vs research in other fields

4 M ★★★ PYQ 2024 · Q1c · textbook

Basis	Engineering	Other fields
Goal	Solve a practical problem; build a product, process or system	Understand nature, society or behaviour
Method	Modelling, simulation, prototyping, testing	Observation, surveys, interviews, case studies
Output	Designs, prototypes, patents, standards	Theories, interpretations, papers
Judged by	Performance, cost, safety, reliability	Truth, explanatory power
Constraints	Budget, safety, regulations, time to market	Mostly ethical and methodological

Good Mechanics Often Join Companies

Goal · Method · Output · Judged by · Constraints

Knowledge flow through citations

4 M ★ Slides 39-40

- Knowledge flows through verbal communication, books, documents, video, audio and images.
- In engineering it flows mainly through books, theses, articles, patents and reports.
- Citing a source transmits knowledge from previous work to an innovation.
- Production of knowledge is linked to the citation network.
- It flows **between co-authors** during collaboration, **among researchers** through citation networks, and **between institutions, departments and fields**.

Big Tigers Always Pounce Randomly

Books · **T**heses · **A**rticles · **P**atents · **R**eports

Acknowledgments and attributions

4 M ★ Slides 43-46

- A brief appreciation of a person, organisation or funding body that contributed to the work. It shows the relationship among people, agencies, institutions and research.
- **What to acknowledge:** quotations, ideas, facts, paraphrasing, the funding organisation, oral discussion or support, and laboratory and computer work.
- **Where:** at the end of the text or as a footnote. In a thesis, immediately after the table of contents.
- **Acknowledgment vs dedication:** acknowledgment thanks those who *helped*; a dedication honours anyone (parents, friends, God). Dedications appear only in books and theses, never in journal papers or patents.

Quiet Indians Find Peace From Old Libraries

Quotations · **I**deas · **F**acts · **P**araphrasing · **F**unding organisation · **O**ral discussion or support · **L**aboratory and computer work

UNIT II · CO2

Research Design & Experimental Designs

Need for research design · Variables · Hypothesis · Experimental and control groups · Treatments · CRD · RBD · Latin Square · Factorial

51 slides

3 of 9 parts in 2024 (17 marks)

Most scoring definitions + drawable designs

Q1 UNIT II

Define: variable, independent variable, dependent variable, extraneous variable, confounded variable, continuous and discrete variables.

5 M · ★★★ Banker · PYQ 2024 · Q1b · Slides 5-10 + Kothari

Variable: any characteristic which is subject to change and can have more than one value (*e.g.* intelligence, motivation, age, gender).

Independent variable (IV): the factor the researcher manipulates or controls to observe its effect: the "cause". It comes in two types: *experimental IV*, directly manipulated (dosage of a drug), and *subject variable*, a characteristic that can't be manipulated but categorises participants (age, gender).

Dependent variable (DV): the outcome measured in response to changes in the IV: the "effect" (test scores, growth, behavioural response).

Extraneous variable: any variable other than the IV that could influence the DV. It adds noise (temperature, lighting, mood, fatigue, time of day).

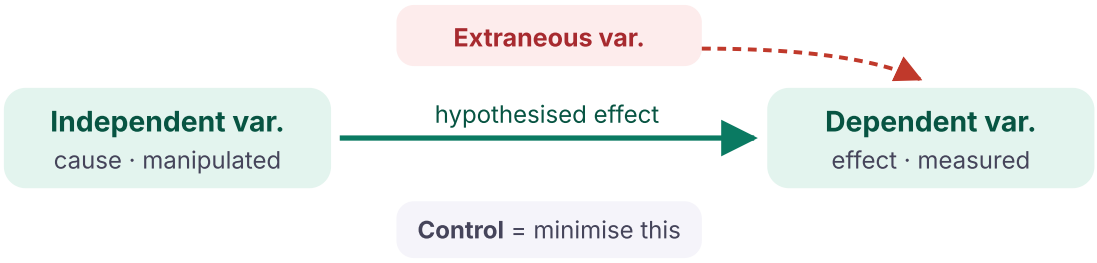
Confounded variable: when an extraneous variable correlates with both the IV and the DV, their separate effects cannot be separated (*e.g.* testing a new teaching method without controlling for students' prior knowledge).

Continuous variable: can take any value in a range, including decimals (age, height, weight).

Discrete variable: takes only whole-number (integer) values (number of children, number of defects).

MEMORY LINE **Very Intelligent Dogs Eat Crunchy Chicken Daily**

Variable · Independent · Dependent · Extraneous · Confounded · Continuous · Discrete



Cause → effect. Control blocks the extraneous variable's influence. If it isn't controlled, the relationship is **confounded**.

L3 TWIST · IDENTIFY THE VARIABLES

"Effect of hours of study on CIE marks." IV: hours of study (continuous). DV: CIE marks. Extraneous: sleep, prior knowledge, difficulty of the paper. Confounding risk: students who study more may also attend more classes.

Q2
UNIT II

What is research design? Explain its need and the features of a good research design.

6 M ★★ Very likely Slides 2-5

Meaning. Once the research problem is defined, preparing the research project is called the "research design". It covers decisions about what, where, when, how much and by what means for an inquiry: a plan or strategy for conducting the research. A good design minimises bias and maximises the reliability of the data, yields maximum information with minimum experimental error, and shows different aspects of a single problem. It depends on the purpose and nature of the problem, so **no single design suits every problem**.

Need for research design

- Smooth sailing of research operations
- Advance planning of the methods
- Great bearing on the reliability of results
- Gives direction
- Helps in decision making
- Prevents blind searching
- Helps anticipate problems in data collection

Sachin And Rahul Don't Drop Perfect Attempts

Smooth sailing · Advance planning · Reliability of results · Direction · Decision making · Prevents blind searching · Anticipate problems

Features of a good design

- The means of obtaining information
- Availability and skills of the researcher and staff
- Availability of time and money
- Flexible enough for exploratory studies
- Accurate with minimum bias for descriptive studies
- Control of extraneous variables
- Statistical correctness for testing the hypothesis

My Aunt Takes Fresh Apples, Carefully Sliced

Means of obtaining information · Availability and skills of researcher · Time and money · Flexible (exploratory) · Accurate, minimum bias (descriptive) · Control of extraneous variables · Statistical correctness

Important concepts in research design: dependent and independent variables, extraneous variable, control, confounded relationship, research hypothesis, experimental and control groups, treatments.

Examiner's eye

Open with an analogy: *"Research design is to research what a blueprint is to a building."* Then two headings (Need, Features), 4–5 points each.

Q3
UNIT II

What is a research hypothesis? State its purpose, and explain null vs alternative and directional vs non-directional hypotheses with examples.

6 M ★★★ Banker Slides 11-16

Definition. A research hypothesis is a predictive statement that relates an independent variable to a dependent variable. It is a proposition or set of propositions set forth as an explanation for some phenomena, either as a provisional conjecture to guide investigation, or accepted as highly probable in the light of established facts.

Purpose of a hypothesis

- Guides and gives direction to the study
- Defines which facts are relevant and not relevant
- Suggests the most appropriate research design
- Provides a framework for organising conclusions
- Limits the research to a specific area
- Offers explanations for relationships that can be empirically tested
- Shows the researcher has enough background knowledge
- Structures the next phase, giving continuity

> MEMORY LINE **Good Dancers Should Feel Light On Stage, Smiling**
> **G**uides and gives direction · **D**efines relevant facts · **S**uggests research design · **F**ramework for conclusions · **L**imits the research · **O**ffers testable explanations · **S**hows background knowledge · **S**tructures the next phase

Types (running example: does a new drug affect blood pressure?)

Type	Meaning	Example
Null (H_0)	No effect, no difference or no relationship. The default position tested against.	H_0: $\mu_1 = \mu_2$
Alternative (H_1 or H_a)	Contradicts H_0: there is an effect. What the researcher aims to support.	H_1: $\mu_1 \neq \mu_2$
Directional	States the expected direction (greater than or less than)	H_1: $\mu_1 < \mu_2$ (drug group has lower BP)
Non-directional	States a difference, without direction	H_1: $\mu_1 \neq \mu_2$

> MEMORY LINE **Never Argue, Don't Nag**
> **N**ull (H0) · **A**lternative (H1) · **D**irectional · **N**on-directional

Classified **by formulation** (null vs alternative) and **by direction** (directional vs non-directional). μ_1 is the mean BP with the drug; μ_2 is the mean BP without it.

> *Examiner's eye*
> H_0 and H_1 sit in **Unit II** slides, so they are in CIE-1 even though "Data Analysis" isn't. Always write the symbols ($\mu_1 = \mu_2$): one line of maths looks like full marks.

Q4 UNIT II

State the principles of experimental design. Explain the Completely Randomised Design (CRD) with an example.

6 M · ★★ Very likely · Slides 25-32

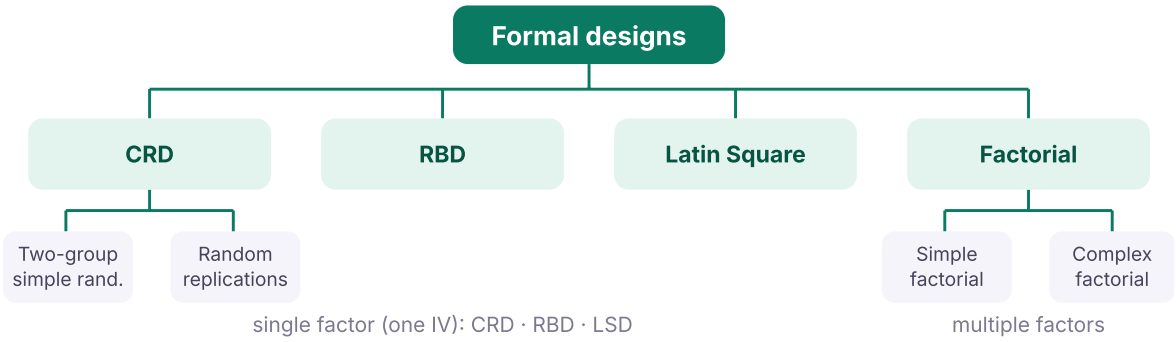
Experimental design methods help the experimenter understand and evaluate the factors that influence a system using statistical approaches.

Principles of experimental design

- **Control:** keep conditions as similar as possible for all treatment groups, apart from the treatments themselves. Then any difference is due to the treatment.
- **Randomisation:** use chance to assign units to treatments. This balances the effect of uncontrolled ("lurking") variables.
- **Replication:** repeat the experiment on many subjects or locations.

MEMORY LINE **Cats Run Rapidly**

Control · Randomisation · Replication



MEMORY LINE **Cool Rabbits Love Food**

CRD · RBD · LSD (Latin square) · Factorial

Completely Randomised Design (CRD)

Treatments are assigned to experimental units completely at random, so each unit has an equal chance of receiving any treatment and the treatment effects can be assessed without bias. Statistical model: $Y_{ij} = \mu + T_i + e_{ij}$ (observation = overall mean + treatment effect + random error).

SLIDE EXAMPLE · BAKING POWDER

Hypothesis: adding more baking powder increases cake height. **Treatments:** ¼, ½, ¾ and 1 tbsp. Cakes are assigned to the four levels **completely at random**, baked, and their heights compared.

Characteristics and advantages

- Random assignment
- Single-factor focus
- Simplicity of layout and analysis
- Maximum degrees of freedom for error
- Flexible: any number of treatments and replications

Rohit Smashes Sixes, Mesmerises Fans

Random assignment · Single-factor focus · Simplicity · Maximum degrees of freedom for error · Flexible

Limitations

- Needs homogeneous units
- Less accurate in field experiments
- Can't detect interactions between factors

Hot Fresh Idli

Homogeneous units needed · Field experiments: less accurate · Interactions not detected

Used in: laboratory experiments, controlled agricultural tests, clinical trials, basic biology and chemistry.

Q5
UNIT II

Briefly explain a randomised block design. Design an RBD to study the effect of three new fertilisers (F1, F2, F3) on rice, wheat and jowar, with yield as the outcome.

6 M · ★★★ Banker · PYQ 2024 · Q3b · Slides 33-37

RBD groups experimental units into blocks based on a characteristic expected to affect the outcome. Within each block, treatments are assigned randomly, which reduces the effect of variability between blocks. Model: $Y_{ij} = \mu + T_j + B_i + E_{ij}$ (overall mean + treatment effect + block effect + error).

- **Blocking:** units with similar characteristics (age, location, crop) form a block, which controls a known nuisance factor.
- **Random assignment within each block.**
- **Complete treatment application:** every treatment appears in every block.

MEMORY LINE **Big Red Car**

Blocking · **R**andom assignment within each block · **C**omplete treatment application

The design for the question

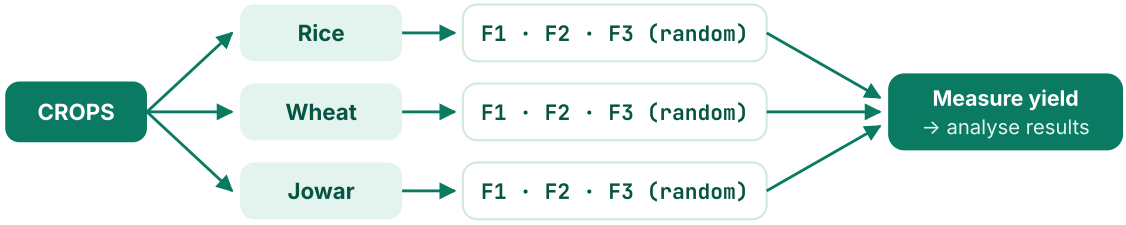
Treatments	F1, F2, F3 (the three new fertilisers)
Blocks	Crop type: Rice, Wheat, Jowar (the nuisance factor)
Experimental unit	One plot of land
Response (DV)	Yield (kg per plot)

MEMORY LINE **Tigers Bite Unless Respected**

Treatments · **B**locks · **U**nit (experimental unit) · **R**esponse (DV)

	Plot 1	Plot 2	Plot 3
Block I · Rice	F2	F3	F1
Block II · Wheat	F1	F3	F2
Block III · Jowar	F3	F2	F1

1. Divide each crop block into **3 plots**.
2. **Randomly** allot F1, F2 and F3 to the plots within each block (by lottery), giving 3 × 3 = **9 plots**.
3. Grow the crops and **measure the yield** of each plot.
4. Compare the mean yield of F1, F2 and F3. The **crop (block) effect is separated out** (two-way ANOVA).



Draw this flow (it matches slide 33) or the field grid above. Drawing both is safest.

Advantages

- Reduces variability, so results are more precise
- Increased statistical power (lower error variance)
- Flexible: agriculture, medicine, industry

Limitations

- Choosing the blocking factor is complex and needs prior knowledge
- Assumes within-block variation is less than between-block variation
- Controls only known nuisance factors

Real Paneer Feast

Reduces variability · **P**ower (increased statistical power) · **F**lexible

Chai Has Kick

Complex to choose blocks · **H**omogeneity assumption · **K**nown nuisance factors only

Examiner's eye

Before drawing, write the four labels: **treatments, blocks, unit, response**. That's 2 marks before you've drawn anything. If they change the crops or fertilisers again, the method stays the same. You can add a control plot **NF** (no fertiliser) like the slide does, to show understanding.

Q6
UNIT II

Explain the Latin Square Design with an example and layout. State its merits and limitations.

6 M | ★★★ Top candidate | Not asked in 2024 | Slides 38–45

LSD is used to control two blocking factors while studying one treatment factor. It is an n × n grid in which each treatment appears exactly once in each row and once in each column. Rows represent one blocking factor and columns the other. Rule: number of treatments = number of rows = number of columns. Model: $Y_{ijk} = \mu + T_j + R_i + C_k + E_{ijk}$ (treatment, row, column effects + error).

SLIDE EXAMPLE · COLD PROTECTION OF STRAWBERRIES

Treatments: 3 irrigation methods: **drip**, **overhead sprinkler**, **none**. **Aim:** find which best protects strawberries from extreme cold. **Extraneous variation: soil moisture** (changes with distance from the canal) and **soil nitrogen** (high to low). **Response:** weight of frozen fruit (lower = more protection).

	N high	N medium	N low
Moisture 1	none	drip	over
Moisture 2	drip	over	none
Moisture 3	over	none	drip

Rows = moisture gradient · columns = nitrogen gradient · each method once per row and once per column

Standard 4 × 4 (A–D = treatments)

A	B	C	D
B	C	D	A
C	D	A	B
D	A	B	C

Slide 41 shows two layouts and asks "which design will best account for both moisture and nitrogen gradients?" The answer is the one where **no treatment repeats in any row or column**: the Latin square.

Merits

- Controls two sources of variability (rows and columns), so error is smaller
- Efficient use of resources: fewer units than separate experiments
- Flexible: agriculture, psychology, clinical studies
- Equal replication gives balanced comparisons

Cute Elephants Fly Balloons
Controls two sources of variability · Efficient use of resources · Flexible · Balanced comparisons

Limitations

- Limited number of treatments: must equal rows and columns
- More complex to design than CRD or RBD
- Assumes minimal variation within rows and columns; otherwise the results are biased

Let's Chill Always
Limited number of treatments · Complex to design · Assumes minimal variation

Examiner's eye

The predicted twist: "Design an LSD for 4 fertilisers with soil fertility varying in two directions." Draw the standard 4 × 4, name the rows and columns, and tick the rule (each letter once per row and column). Practise drawing it in 60 seconds.

Q7
UNIT II

What is a factorial design? Explain simple and complex factorial designs. List all combinations of a 3×2×2 design (therapy type × duration × therapist gender).

6 M | ★★★ Banker | PYQ 2024 · Q2b | Slides 46-50

A **factorial design** evaluates the effects of two or more independent variables (factors) on a dependent variable. It measures the main effects (each factor's individual impact) and the interaction effects (how one factor's effect changes with the level of another). **Number of combinations = product of the levels**.

Simple factorial: two factors, each at two levels (2 × 2). Example: drug dosage (low, high) × duration (short, long) gives **4 combinations**: low-short, low-long, high-short, high-long.

Complex factorial: more than two factors, or factors with more than two levels (2 × 3, 3 × 2 × 2). Example: **3 × 2 × 2 = 12 combinations**:

Therapy ↓ · Duration / Gender →	Short · Male	Short · Female	Long · Male	Long · Female
Cognitive Behavioural (CBT)	1	2	3	4
Psychodynamic	5	6	7	8
Humanistic	9	10	11	12

In the exam, also write them as a list: (1) CBT, short, male · (2) CBT, short, female · (3) CBT, long, male · (4) CBT, long, female · (5) Psychodynamic, short, male · (6) Psychodynamic, short, female · (7) Psychodynamic, long, male · (8) Psychodynamic, long, female · (9) Humanistic, short, male · (10) Humanistic, short, female · (11) Humanistic, long, male · (12) Humanistic, long, female.

Advantages

- Studies the combined effect of several factors simultaneously: economical and efficient
- Wide range of combinations helps predict what happens when factors combine
- Saves time and material compared with separate experiments
- Avoids the environmental-change problems of single-factor experiments
- More complete information: evaluates interaction effects

Coffee With Sugar Energises Instantly
Combined effect of factors · Wide range of combinations · Saves time and material · Environmental-change problems avoided · Interaction effects evaluated

Disadvantages

- Execution and analysis get complex as combinations grow
- You may be forced to include uninteresting combinations, wasting time and material
- More factors means bigger blocks and more heterogeneity, so more error and less precision

Chicken Fried Biryani
Complex execution and analysis · Forced to include uninteresting combinations · Bigger blocks, more error

Examiner's eye
Use a **tree** (3 branches, then 2, then 2) or the grid above so you never miss a combination. Write "3 × 2 × 2 = 12" first: the examiner sees you know the count.

Rapid-fire • Unit II

Compact answers for 3–5 mark parts

Treatments

4 M ★★ Slides 17–20

- **Treatment** is what is administered to participants or units: a drug, therapy, programme or other intervention expected to have an effect.
- **Experimental treatments** are the new interventions being tested. **Control treatments** are a standard treatment, a placebo or no treatment.
- **Purpose:** to find the treatment's effect on the DV by comparing with the control.
- **Design:** plan the dosage, duration and delivery. Random assignment reduces bias. Measure pretest and posttest.
- *Example:* hypertension trial: the treatment group gets the new drug, the control group a placebo; compare BP before and after.

Experimental group vs control group

4 M ★★ Slides 21–24

Basis	Experimental	Control
Treatment	Receives the treatment	No treatment, a placebo or the standard one
Role	Shows the IV's effect	Baseline for comparison
Example	Gets the actual drug	Gets a placebo pill

The groups should be identical except for the treatment. **Why both:** they establish causality, reduce bias and variability, and benchmark results.

Cows Remain Beautiful

Causality established · Reduce bias and variability · Benchmark results

Compare CRD, RBD and LSD

4–5 M ★★ Slides 28–45

Basis	CRD	RBD	LSD
Blocking factors controlled	0	1 (blocks)	2 (rows and columns)
Allocation	Completely at random	Random within each block	Once per row and once per column
Model	$Y = \mu + T + e$	$Y = \mu + T + B + E$	$Y = \mu + T + R + C + E$
Best when	Units are homogeneous (lab)	One known nuisance factor	Two known nuisance gradients (field)
Main limit	Poor for field experiments	Only known factors controlled	Treatments = rows = columns

Cause and effect, and why variables matter

3 M ★ Slides 7, 10

- The **cause** is *why* something happens (the IV); the **effect** is *what* happens (the DV).
- Variables matter for hypothesis development (clear variables give testable hypotheses), experimental design (knowing how to manipulate and measure) and data analysis (variable type decides the statistics).

Happy Elephants Dance

Hypothesis development · Experimental design · Data analysis

Why randomise? Why replicate?

3 M ★ Slide 26

- **Randomisation** creates roughly equivalent groups by balancing lurking variables, which removes the researcher's selection bias.
- **Replication** lets you estimate experimental error and makes results reliable, not one-off.
- **Control** makes sure an observed difference is due to the treatment.

UNIT III · CO3 · SLIDES 1–54 ONLY

Data Collection, Sampling & Scaling

Primary and secondary data · Observation, interview, questionnaire, schedule · Sampling fundamentals · Sample design · Measurement scales · Rating scale

54
slides in scope (of 81)

New
not in the 2024 CIE-1

~3 parts
expected this year

Q1
UNIT III

Differentiate between primary and secondary data. What characteristics must secondary data possess before a researcher uses it?

6 M · ★★★ Banker · Slides 3–5, 17–18

Data collection begins after the research problem has been defined and the research design chalked out. There are two types of data.

Basis	Primary data	Secondary data
Meaning	Collected afresh and for the first time; original in character	Already collected by someone else and passed through the statistical process
Collected by	The researcher, for this study	Other agencies, for some other purpose
Methods and sources	Observation, interview, questionnaire, schedule; experiments; surveys	Published (government reports, journals, books) and unpublished (diaries, letters) sources
Cost and time	High	Low
Suitability	Fits the purpose exactly	Must be checked before use
Example	Your own survey of 200 students	Census of India figures

MEMORY LINE **Mom Cooks Masala Chai Super Early**
Meaning · **C**ollected by · **M**ethods and sources · **C**ost and time · **S**uitability · **E**xample

Characteristics secondary data must possess: R · S · A

1. Reliability. Check: who collected the data? What were the sources? Were proper methods used? When was it collected? Was there bias of the compiler? What accuracy was desired, and was it achieved?

2. Suitability. Data suitable for one enquiry may not be suitable for another. Unsuitable data should not be used.

3. Adequacy. If the level of accuracy is inadequate for the present enquiry, the data should not be used.

MEMORY LINE **Rahul Scores Always**
Reliable · **S**uitable · **A**dequate

MEMORY LINE **Why So Many Teachers Bunk Again**

Who collected it? · **S**ources? · **M**ethods proper? · **T**ime of collection? · **B**ias of compiler? · **A**ccuracy achieved?

Published sources: central, state and local government publications; foreign governments and international bodies; technical and trade journals; books, magazines and newspapers; reports of business associations, banks and stock exchanges; reports by research scholars and universities; public records and historical documents. **Unpublished sources:** diaries, letters, unpublished biographies and autobiographies, and records held by scholars, trade associations and labour bureaus.

Q2
UNIT III

Explain the observation and interview methods of collecting primary data, with their advantages and limitations.

6 M · ★★★ Banker · Slides 5-11

The four methods of collecting primary data are observation, interview, questionnaire and schedule.

MEMORY LINE **Only Idiots Question Sachin**

Observation · **I**nterview · **Q**uestionnaire · **S**chedule

(i) Observation method

Information is obtained by the investigator's own direct observation, without asking the respondent. It becomes a scientific tool when it serves a formulated research purpose, is systematically planned and recorded, and is subjected to checks and controls on validity and reliability. Kinds: structured or unstructured, participant or non-participant, controlled or uncontrolled.

Advantages

- Subjective bias is eliminated
- Information relates to what is currently happening
- Independent of respondents' willingness to respond
- Works with respondents unable to give verbal reports

Students **C**an **I**gnore **N**onsense

Subjective bias eliminated · **C**urrent happenings · **I**ndependent of willingness · **N**on-verbal respondents OK

Limitations

- Expensive
- Limited information
- Unforeseen factors may interfere

Eat **L**ess **U**nhealthy

Expensive · **L**imited information · **U**nforeseen factors

(ii) Interview method

Presentation of oral-verbal stimuli and reply in terms of oral-verbal responses. Done through **personal interviews** (face to face) or **telephone interviews**.

- **Structured:** a set of predetermined questions with highly standardised recording; the interviewer follows a rigid procedure, in a prescribed form and order.
- **Unstructured:** flexible approach; can ask supplementary questions, omit questions or change their order. Less comparable between interviews, and harder to analyse.

Advantages

- More information, in greater depth
- The interviewer can overcome resistance
- Greater flexibility
- Personal information obtained easily
- Samples controlled more effectively
- Language adapted to the respondent's level
- Can capture spontaneous reactions

My Old Granny Prefers Sweet Lassi, Seriously

More information, in depth · Overcome resistance · Greater flexibility · Personal information · Samples controlled · Language adapted · Spontaneous reactions

Limitations

- Very expensive for wide samples
- Bias of interviewer and respondent
- Important officials hard to approach
- Time-consuming
- The interviewer's presence may over-stimulate the respondent
- Selecting and training staff is complex
- Needs proper rapport

Every Boy Asks The Others' Secret Recipe

Expensive · Bias (interviewer and respondent) · Approach to officials is hard · Time-consuming · Over-stimulates respondent · Staff selection and training complex · Rapport needed

---

Q3
UNIT III

What are questionnaires and schedules? Distinguish between them, and state the merits and limitations of questionnaires.

6 M · ★★★ Banker · Slides 12-16

Questionnaire: a number of questions printed or typed in a definite order on a form, mailed to respondents, who read, understand and answer on their own. Popular for big enquiries by individuals, organisations and governments.

Schedule: a proforma containing a set of questions, filled in by enumerators specially appointed for the purpose. They ask the questions in order, record the replies, and explain the aims and remove difficulties. Enumerators must be carefully selected and trained, intelligent, able to cross-examine, honest and patient. Population censuses are conducted this way.

Basis	Questionnaire	Schedule
Filled by	The respondent, sent by mail	The enumerator, who asks the respondent
Cost	Cheap	Expensive (enumerators' pay and training)
Non-response	High: low return rate	Low: the enumerator gets answers
Respondent	Must be literate and cooperative	Can be used even with illiterate respondents
Personal contact	None	Direct, face to face
Speed	Slow (waiting for returns)	Collected in time
Accuracy	Risk of ambiguous or omitted replies	More complete; depends on enumerators' honesty
Coverage	Very wide; large samples possible	Limited by the number of enumerators

MEMORY LINE **Five Cute Ninjas Run Past Seven Angry Cops**

Filled by · Cost · Non-response · Respondent · Personal contact · Speed · Accuracy · Coverage

Merits of a questionnaire

- Low cost
- Free from interviewer bias; answers in the respondents' own words
- Adequate time for well-thought-out answers
- Reaches respondents who are not easily approachable
- Large samples, so more dependable results

Lazy Boys Take A Leave

Low cost · Bias of interviewer absent · Time to think · Approach the unapproachable · Large samples

Limitations

- Low rate of return
- Respondents must be educated and cooperating
- Control is lost once it's sent; hard to amend
- Ambiguous or omitted replies
- Hard to know if respondents are representative
- Slow

Lions Eat Chicken And Rice Slowly

Low return rate · Educated respondents only · Control lost after sending · Ambiguous or omitted replies · Representative? hard to know · Slow

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Q4
UNIT III

What is sampling? Why is it needed? Define universe, sampling frame, sample design, statistic and parameter, sampling error, precision, confidence level and significance level.

6 M · ★★★ Banker · Slides 20–26

A complete enumeration of all items in the population is a census inquiry. **Sampling** is the selection of some part of an aggregate or totality on the basis of which a judgement or inference about the aggregate is made: obtaining information about the whole population by examining only part of it.

Need for sampling

1. Saves time and money.
2. Can give more accurate measurements when done by trained investigators.
3. The only way when the population is infinite.
4. The only choice for destructive testing (for example, the life of light bulbs).
5. Lets you estimate the sampling errors.

MEMORY LINE **Tea And Idli Daily, Enjoy**

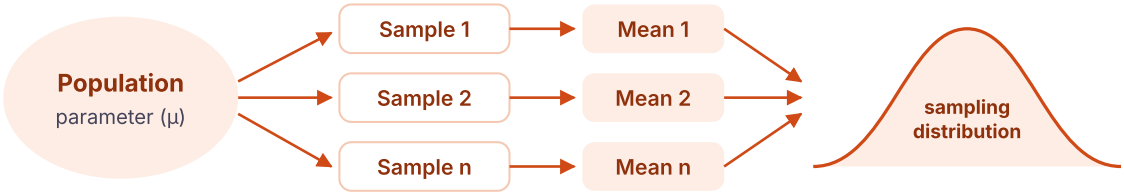
Time and money saved · Accuracy · Infinite population · Destructive testing · Estimate sampling error

Important definitions

Universe or population	The universe is the total of items in any field of inquiry; the population is the total of items about which information is desired.
Sampling frame	A list of all sampling units from which the sample is drawn (for example, a telephone directory for a city opinion survey). It should represent the population well.
Sample design	A definite plan for obtaining a sample from the sampling frame.
Statistic vs parameter	A statistic describes a sample (sample mean); a parameter describes the population (population mean).
Sampling error	The inaccuracy that comes from studying only part of the population (also called error variance).
Precision	The range within which the population value will lie. For an estimate of ₹4000 with ±4% precision, the true value is **₹3840 to ₹4160**.
Confidence and significance level	Confidence: the expected % of times the actual value falls within the precision limits. At 95% confidence, the significance level is 100 − 95 = 5%.
Sampling distribution	The means (or other statistics) of many samples, with their frequencies. Its mean equals the population mean.

MEMORY LINE **Uncle Fred Drives Six Expensive Porsches Carefully Daily**

Universe / population · **F**rame (sampling frame) · **D**esign (sample design) · **S**tatistic vs parameter · **E**rror (sampling error) · **P**recision · **C**onfidence and significance level · **D**istribution (sampling distribution)



Q5 UNIT III

Explain the steps in sample design and the criteria for selecting a sampling procedure.

6 M ★★ Very likely Slides 27-29

Steps in sample design

1 · Type of universe	Finite or infinite
2 · Sampling unit	Geographical (state, village), construction (house, flat), social (family, school) or an individual
3 · Source list	The sampling frame the sample is drawn from
4 · Size of sample	The number of items selected from the universe
5 · Parameters of interest	The population values wanted (for example, the average height of people)
6 · Budgetary constraint	The money available limits the size and type of sample
7 · Sampling procedure	The technique used to select the items

MEMORY LINE **Unlucky Uncle Sold Seven Pens By Post**

Universe type · **U**nit (sampling unit) · **S**ource list (frame) · **S**ize of sample · **P**arameters of interest · **B**udgetary constraint · **P**rocedure (sampling procedure)

Criteria for selecting a sampling procedure

Sampling involves two costs: the cost of collecting data and the cost of an incorrect inference. Wrong inferences come from two causes: systematic bias and sampling error.

Causes of systematic bias:

1. Inappropriate sampling frame: a biased representation of the universe
2. Defective measuring device: constantly in error
3. Non-respondents: not all selected individuals can be sampled
4. Indeterminacy principle: people behave differently when observed
5. Natural bias in reporting data by respondents

MEMORY LINE **I Do Not Invite Nonsense**

Inappropriate sampling frame · **D**efective measuring device · **N**on-respondents · **I**ndeterminacy principle · **N**atural bias in reporting

Sampling error is the random variation of sample estimates around the true population parameter. Measuring it gives the precision of the sampling plan. Increasing the sample size improves precision, but it also raises cost and enhances systematic bias. The effective fix is a better sampling design with smaller error for the same size and cost.

Systematic bias	Sampling error
Caused by errors in the **sampling procedure**	Caused by **random chance**
Not reduced by a bigger sample	Reduced by a bigger sample or a better design

---

Q6
UNIT III

State the characteristics of a good sample design. Explain the types of sample design, distinguishing probability from non-probability sampling.

6 M · ★★★ Banker · Slides 38–43

Characteristics of a good sample design

1. Results in a truly representative sample
2. Results in a small sampling error
3. Is viable within the funds available
4. Lets systematic bias be controlled better
5. Results can be applied to the universe with a reasonable level of confidence

MEMORY LINE **Rahul Should Visit Chennai Again**

Representative · **S**mall sampling error · **V**iable within funds · **C**ontrols systematic bias · **A**pplies to the universe with confidence

Types of sample design

Sample designs are classified on two bases: the representation basis (probability or non-probability) and the element selection technique (unrestricted, where each element is drawn individually from the population at large, or restricted, meaning all other forms).

Element selection ↓ · Representation →	Probability sampling	Non-probability sampling
Unrestricted	Simple random sampling	Haphazard or convenience sampling
Restricted	Complex random sampling: cluster, systematic, stratified, etc.	Purposive sampling: quota, judgement

MEMORY LINE **Sweet Chai Cools People**

Simple random (probability, unrestricted) · **C**omplex random (probability, restricted) · **C**onvenience (non-probability, unrestricted) · **P**urposive (non-probability, restricted)

Non-probability sampling (also called deliberate, purposive or judgement sampling): there is no basis for estimating the probability of each item being included. Items are selected deliberately by the researcher, whose choice is supreme. *Example:* a few towns and villages chosen to represent a whole state. Risk: the personal element (organiser's bias) enters, and sampling error cannot be estimated. It suits small inquiries because it saves time and money. **Quota sampling** is an example: interviewers fill quotas from different strata at their own discretion. It is cheap and convenient, but not random.

Probability sampling (also called random or chance sampling): every item of the universe has an equal chance of inclusion, like a lottery, so blind chance alone decides. Errors of estimation can be measured, and it obeys the law of statistical regularity: a random sample has the same composition as the universe. That makes it the best technique for a representative sample. *Without replacement*, a selected item can't appear again; *with replacement*, it is returned and could appear twice.

Basis	Probability	Non-probability
Selection	By chance (lottery)	Deliberate, by judgement
Chance of inclusion	Equal and known	Unknown
Sampling error	Can be measured	Cannot be estimated
Bias	Free from personal bias	Organiser's bias likely
Cost	Higher	Lower; convenient

MEMORY LINE **Some Cats Sleep Beside Cows**

Selection · **C**hance of inclusion · **S**ampling error · **B**ias · **C**ost

Extra one-liners (Kothari): *simple random* means every item has an equal chance · *systematic* means every k-th item from a list · *stratified* means dividing into homogeneous strata and sampling each · *cluster or area* means sampling whole groups · *multi-stage* means sampling in stages (state, district, village).

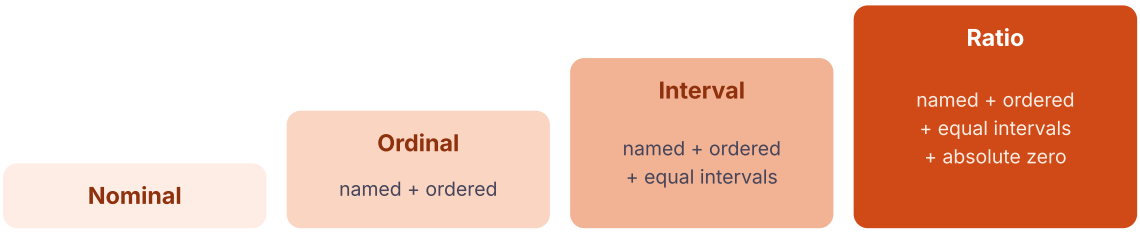
Q7
UNIT III

What is measurement? Explain the four levels of measurement with examples, and the sources of error in measurement.

6 M · ★★★ Banker · Slides 44–52

Measurement is the process of assigning numbers to objects or observations: mapping aspects of a domain onto aspects of a range according to some rule. We measure physical objects as well as abstract concepts.

Scale	What it does	Examples	Allowed statistics
Nominal	Labels variables into distinct categories, with no quantitative value or order (also called the categorical scale)	Gender (M/F), political preference, place of residence, blood group	Counts, mode
Ordinal	Shows the order of variables, not the difference between them	Likert scale: very unsatisfied (1) to very satisfied (5); ranks	Median, percentiles
Interval	Order plus equal, known differences, but no true zero	Temperature: 80° − 50° = 70° − 40°	Mean, SD
Ratio	Order, known differences plus a true (absolute) zero	Height, weight, income, age	All, including ratios



Each level keeps everything below it and adds one property (slide 51).

MEMORY LINE **No One Is Right**
Nominal · Ordinal · Interval · Ratio

Sources of error in measurement

1. **Respondent:** reluctant to express strong negative feelings, or knows little but won't admit ignorance.
2. **Situation:** any condition that strains the interview, such as someone else being present.
3. **Measurer:** the interviewer distorts responses by rewording or reordering questions.
4. **Instrument:** a defective measuring instrument.

MEMORY LINE **Rats Steal My Idli**
Respondent · Situation · Measurer · Instrument

Examiner's eye
The examiner loves "which scale is this?" Practise with the drills on page 29. The giveaway: **can you say "twice as much"?** If yes, it's ratio. If 0 doesn't mean "none" (0 °C), it's interval.

Rapid-fire • Unit III

Compact answers for 3–5 mark parts

Important sampling distributions

5 M ★★ Slides 30-35

1. **Of the mean:** the distribution of means of all possible random samples of a given size. It is normal if the population is normal, and near-normal for large samples.
2. **Of the proportion:** for attribute data (defective or non-defective). It is binomial, becoming normal as n grows.
3. **Student's t:** used when the population SD is unknown and n < 30. Symmetrical; approaches normal as n grows.
4. **F:** compares the variances of two samples from two normal populations with the same variance.
5. **Chi-square:** not symmetrical, all values positive. Used for observed vs expected frequencies and goodness of fit.

My Parents Took Five Chocolates

Mean · Proportion · t (Student's t) · F · Chi-square

Central Limit Theorem and Standard Error

4 M ★★ Slides 36-37

CLT: whether or not the population is normal, the means of samples drawn from it follow a normal distribution when n > 30. If the population is normal, sample means are normal for any n.

Standard error (SE): the standard deviation of the sampling distribution of a statistic. For the mean, SE = σ / √n. It:

- tests whether a difference could be due to chance
- gives an idea of the reliability and precision of a sample
- sets the limits within which the population parameter is expected to lie

Cats Really Love

Chance: tests if a difference is due to chance · Reliability and precision of the sample · Limits for the population parameter

Meaning of scaling and the rating scale

4 M ★★ Slides 53-54 (last CIE-1 slide)

Scaling means assigning numbers to various degrees of opinion, attitude and other concepts. It is done in two ways: (i) judging some characteristic of an individual, or (ii) constructing questionnaires.

Rating scale: a qualitative description of a limited number of aspects of a thing or traits of a person. The object is judged in absolute terms against specified criteria. Its results compare favourably with other methods, it takes less time, is interesting to use, has a wide range of applications, and works with many variables.

Cool Lions In White Masks

Compares favourably with other methods · Less time · Interesting to use · Wide range of applications · Many variables

Example (graphic rating): "How do you like the product?" Like very much · Like somewhat · Neutral · Dislike somewhat · Dislike very much.

Structured vs unstructured interview

3 M ★ Slides 8-9

Basis	Structured	Unstructured
Questions	Predetermined, fixed order	Flexible; can add, omit or reorder
Recording	Highly standardised	Not standardised
Comparability	High; easy analysis	Low; analysis hard and slow
Interviewer freedom	Rigid procedure	Much greater freedom

Queens Rarely Cook Fish

Questions · Recording · Comparability · Freedom of interviewer

Scenario drills: the "apply" marks

Last year 19 of the 45 marks on the paper were L3. Cover the answer, think for 30 seconds, then check.

#	Scenario	Answer
1	A study tests whether background music affects students' typing speed. Identify the IV, DV and extraneous variables.	**IV:** music (on or off, or type of music) · **DV:** typing speed (wpm) · **Extraneous:** typing experience, keyboard type, fatigue, time of day
2	List all combinations: teaching mode (online, offline) × class size (small, medium, large).	**2 × 3 = 6:** online-small, online-medium, online-large, offline-small, offline-medium, offline-large (a complex factorial)
3	Design a layout for 4 fertilisers A–D on a field whose fertility varies east to west and north to south.	**Latin square 4 × 4.** Rows control the north–south gradient, columns control the east–west gradient; each letter once per row and once per column (see Unit II · Q6)
4	Compare 3 diets on rats of 3 different age groups. Which design?	**RBD.** Age group = block; diets assigned randomly within each age block
5	Study how temperature AND catalyst together affect reaction yield, including their interaction. Which design?	**Factorial design.** Only it measures **interaction effects**
6	Compare 4 training methods on very similar employees in a controlled lab. Which design?	**CRD.** Units are homogeneous, so assign completely at random
7	Write H_0 and a directional H_1: "Students who use flashcards score higher than those who don't."	**H_0: $\mu_1 = \mu_2$ · H_1: $\mu_1 > \mu_2$** (directional, where μ_1 is the flashcard group)
8	Which scale? Blood group · class rank · temperature in °C · monthly salary · satisfaction on a 1–5 scale.	**Nominal** · **Ordinal** · **Interval** · **Ratio** · **Ordinal** (Likert)
9	Testing the life of light bulbs: census or sample? And picking every 10th bolt off a production line is which method?	**Sample:** destructive testing leaves no choice · **Systematic sampling** (probability, restricted)
10	You use Census of India data for your project. What type of data is this, and what must you check?	**Secondary data.** Check **reliability, suitability and adequacy** (who collected it, method, time, bias, accuracy)

Self-test: write these in 25 minutes (a full 15-mark question)

Mock A

(a) Explain the types of research. **6M**
(b) Explain LSD with a layout. **5M**
(c) Differentiate questionnaire and schedule. **4M**

Mock B

(a) Explain the functions of citation. **6M**
(b) Define the types of variables with an example. **5M**
(c) Explain the four measurement scales. **4M**

Mock C

(a) Explain the need for sampling and its key terms. **6M**
(b) Design an RBD for a given case. **5M**
(c) Compare critical and creative reading. **4M**

Memory lines: first letter = the answer

All 64 memory lines from this guide in one place. Say the sentence, then turn each first letter back into the answer. To test yourself, cover the right-hand column.

Unit I · Research Methodology

What to remember	Memory line	Expands to
Types of research (4 pairs)	**D**ad **A**lways **A**sks **F**or **Q**uick **Q**uestions, **C**lear **E**xplanations	**D**escriptive · **A**nalytical · **A**pplied · **F**undamental · **Q**uantitative · **Q**ualitative · **C**onceptual · **E**mpirical
Other types of research	**O**nly **L**azy **F**olks **L**ove **S**leeping, **C**offee **C**ures **D**rowsiness	**O**ne-time · **L**ongitudinal · **F**ield-setting · **L**aboratory · **S**imulation · **C**linical / diagnostic · **C**onclusion-oriented · **D**ecision-oriented
Research comprises (process)	**D**o **H**ave **C**offee **D**uring **T**ests	**D**efine the problem · **H**ypothesis · **C**ollect, organise and evaluate data · **D**eductions · **T**est the conclusions
Objectives of research	**E**very **D**ay **D**reams **H**appen	**E**xploratory (familiarity, new insights) · **D**escriptive (portray characteristics) · **D**iagnostic (frequency, association) · **H**ypothesis-testing (causal)
Motivation in research	**D**egree **C**an **I**nspire **S**ome **R**espect	**D**egree and its benefits · **C**hallenge of unsolved problems · **I**ntellectual joy · **S**ervice to society · **R**espectability
Why ethics is important	**I** **T**rust **P**arents' **C**ommon **A**dvice	**I**ntegrity of research · **T**rust and credibility · **P**revents harm · **C**ollaboration · **A**ccountability
Ethical principles (7)	**H**onest **O**fficers **I**n **C**ourt **O**penly **R**espect **C**onfidentiality	**H**onesty · **O**bjectivity · **I**ntegrity · **C**arefulness · **O**penness · **R**espect for IP · **C**onfidentiality
Types of research misconduct	**F**ake **F**acts **P**unished	**F**abrication · **F**alsification · **P**lagiarism
Forms of plagiarism	**D**on't **S**teal **M**y **P**roject **A**nd **M**arks	**D**irect · **S**elf-plagiarism · **M**osaic · **P**araphrasing without credit · **A**ccidental · **M**isattribution
Functions of citation	**V**ery **A**ccurate **D**ata	**V**erification · **A**cknowledgment · **D**ocumentation
Citations that fail	**S**ome **B**ad **S**cholars **C**heat	**S**purious · **B**iased · **S**elf-citations · **C**oercive
What existing knowledge gives you	**C**ats **S**leep **O**n **T**ables	**C**ontext · **S**ignificance · **O**riginality · **T**ools
What a literature review must do	**S**mart **K**ids **C**an **I**dentify **I**nteresting **R**esearch	**S**ummarise the state of the art · **K**ey concepts and factors · **C**omplementary approaches · **I**nconsistencies and shortcomings · **I**nconclusive or contradictory results · **R**eason for further work
Analysis and synthesis of prior art	**H**ungry **M**onkeys **C**an **C**rack **S**hells	**H**ypothesis · **M**odels and experimental conditions · **C**onnections · **C**ompare and contrast · **S**trong points and loopholes
Conceptualising research	**S**mart **K**ids **W**in	**S**ignificant problem · **K**nowledge that will address it · **W**ay to make that new knowledge
Critical vs creative reading (table rows)	**A**lways **A**sk **Q**uestions, **D**on't **O**verthink	**A**im · **A**ttitude · **Q**uestions asked · **D**ifficulty · **O**utput
Framing a title (6 points)	**I** **T**otally **L**ove **M**om's **K**heer, **G**uys	**I**nformative and accurate · **T**ype of title · **L**ength · **M**arkers · **K**eywords · **G**ains attention
Title: the slide's 3 aspects	**T**ea **L**overs **M**eet	**T**ype of title · **L**ength of title · **M**arkers (colon, hyphen, ?)
Engineering vs other research (table rows)	**G**ood **M**echanics **O**ften **J**oin **C**ompanies	**G**oal · **M**ethod · **O**utput · **J**udged by · **C**onstraints
Knowledge flow in engineering	**B**ig **T**igers **A**lways **P**ounce **R**andomly	**B**ooks · **T**heses · **A**rticles · **P**atents · **R**eports
What to acknowledge	**Q**uiet **I**ndians **F**ind **P**eace **F**rom **O**ld **L**ibraries	**Q**uotations · **I**deas · **F**acts · **P**araphrasing · **F**unding organisation · **O**ral discussion or support · **L**aboratory and computer work

Unit II · Research & Experimental Design

What to remember	Memory line	Expands to
Types of variables	**V**ery **I**ntelligent **D**ogs **E**at **C**runchy **C**hicken **D**aily	**V**ariable · **I**ndependent · **D**ependent · **E**xtraneous · **C**onfounded · **C**ontinuous · **D**iscrete
Need for research design	**S**achin **A**nd **R**ahul **D**on't **D**rop **P**erfect **A**ttempts	**S**mooth sailing · **A**dvance planning · **R**eliability of results · **D**irection · **D**ecision making · **P**revents blind searching · **A**nticipate problems
Features of a good design	**M**y **A**unt **T**akes **F**resh **A**pples, **C**arefully **S**liced	**M**eans of obtaining information · **A**vailability and skills of researcher · **T**ime and money · **F**lexible (exploratory) · **A**ccurate, minimum bias (descriptive) · **C**ontrol of extraneous variables · **S**tatistical correctness
Purpose of a hypothesis	**G**ood **D**ancers **S**hould **F**eel **L**ight **O**n **S**tage, **S**miling	**G**uides and gives direction · **D**efines relevant facts · **S**uggests research design · **F**ramework for conclusions · **L**imits the research · **O**ffers testable explanations · **S**hows background knowledge · **S**tructures the next phase
Types of hypothesis	**N**ever **A**rgue, **D**on't **N**ag	**N**ull (H0) · **A**lternative (H1) · **D**irectional · **N**on-directional
Principles of experimental design	**C**ats **R**un **R**apidly	**C**ontrol · **R**andomisation · **R**eplication
The four formal designs	**C**ool **R**abbits **L**ove **F**ood	**C**RD · **R**BD · **L**SD (Latin square) · **F**actorial
CRD: characteristics and advantages	**R**ohit **S**mashes **S**ixes, **M**esmerises **F**ans	**R**andom assignment · **S**ingle-factor focus · **S**implicity · **M**aximum degrees of freedom for error · **F**lexible
CRD: limitations	**H**ot **F**resh **I**dli	**H**omogeneous units needed · **F**ield experiments: less accurate · **I**nteractions not detected
RBD: key features	**B**ig **R**ed **C**ar	**B**locking · **R**andom assignment within each block · **C**omplete treatment application
RBD: the 4 labels to write first	**T**igers **B**ite **U**nless **R**espected	**T**reatments · **B**locks · **U**nit (experimental unit) · **R**esponse (DV)
RBD: advantages	**R**eal **P**aneer **F**east	**R**educes variability · **P**ower (increased statistical power) · **F**lexible
RBD: limitations	**C**hai **H**as **K**ick	**C**omplex to choose blocks · **H**omogeneity assumption · **K**nown nuisance factors only
LSD: merits	**C**ute **E**lephants **F**ly **B**alloons	**C**ontrols two sources of variability · **E**fficient use of resources · **F**lexible · **B**alanced comparisons
LSD: limitations	**L**et's **C**hill **A**lways	**L**imited number of treatments · **C**omplex to design · **A**ssumes minimal variation
Factorial: advantages	**C**offee **W**ith **S**ugar **E**nergises **I**nstantly	**C**ombined effect of factors · **W**ide range of combinations · **S**aves time and material · **E**nvironmental-change problems avoided · **I**nteraction effects evaluated
Factorial: disadvantages	**C**hicken **F**ried **B**iryani	**C**omplex execution and analysis · **F**orced to include uninteresting combinations · **B**igger blocks, more error
Why experimental + control groups	**C**ows **R**emain **B**eautiful	**C**ausality established · **R**educe bias and variability · **B**enchmark results
Why variables matter	**H**appy **E**lephants **D**ance	**H**ypothesis development · **E**xperimental design · **D**ata analysis

Unit III · Data Collection, Sampling & Scaling

What to remember	Memory line	Expands to
Primary vs secondary (table rows)	**M**om **C**ooks **M**asala **C**hai **S**uper **E**arly	**M**eaning · **C**ollected by · **M**ethods and sources · **C**ost and time · **S**uitability · **E**xample
Secondary data must be	**R**ahul **S**cores **A**lways	**R**eliable · **S**uitable · **A**dequate
Reliability check questions	**W**hy **S**o **M**any **T**eachers **B**unk **A**gain	**W**ho collected it? · **S**ources? · **M**ethods proper? · **T**ime of collection? · **B**ias of compiler? · **A**ccuracy achieved?
Methods of collecting primary data	**O**nly **I**diots **Q**uestion **S**achin	**O**bservation · **I**nterview · **Q**uestionnaire · **S**chedule

Observation: advantages	**S**tudents **C**an **I**gnore **N**onsense	**S**ubjective bias eliminated · **C**urrent happenings · **I**ndependent of willingness · **N**on-verbal respondents OK
Observation: limitations	**E**at **L**ess **U**nhealthy	**E**xpensive · **L**imited information · **U**nforeseen factors
Interview: advantages	**M**y **O**ld **G**ranny **P**refers **S**weet **L**assi, **S**eriously	**M**ore information, in depth · **O**vercome resistance · **G**reater flexibility · **P**ersonal information · **S**amples controlled · **L**anguage adapted · **S**pontaneous reactions
Interview: limitations	**E**very **B**oy **A**sks **T**he **O**thers' **S**ecret **R**ecipe	**E**xpensive · **B**ias (interviewer and respondent) · **A**pproach to officials is hard · **T**ime-consuming · **O**ver-stimulates respondent · **S**taff selection and training complex · **R**apport needed
Questionnaire vs schedule (table rows)	**F**ive **C**ute **N**injas **R**un **P**ast **S**even **A**ngry **C**ops	**F**illed by · **C**ost · **N**on-response · **R**espondent · **P**ersonal contact · **S**peed · **A**ccuracy · **C**overage
Questionnaire: merits	**L**azy **B**oys **T**ake **A** **L**eave	**L**ow cost · **B**ias of interviewer absent · **T**ime to think · **A**pproach the unapproachable · **L**arge samples
Questionnaire: limitations	**L**ions **E**at **C**hicken **A**nd **R**ice **S**lowly	**L**ow return rate · **E**ducated respondents only · **C**ontrol lost after sending · **A**mbiguous or omitted replies · **R**epresentative? hard to know · **S**low
Need for sampling	**T**ea **A**nd **I**dli **D**aily, **E**njoy	**T**ime and money saved · **A**ccuracy · **I**nfinite population · **D**estructive testing · **E**stimate sampling error
Sampling definitions (8 terms)	**U**ncle **F**red **D**rives **S**ix **E**xpensive **P**orsches **C**arefully **D**aily	**U**niverse / population · **F**rame (sampling frame) · **D**esign (sample design) · **S**tatistic vs parameter · **E**rror (sampling error) · **P**recision · **C**onfidence and significance level · **D**istribution (sampling distribution)
Steps in sample design	**U**nlucky **U**ncle **S**old **S**even **P**ens **B**y **P**ost	**U**niverse type · **U**nit (sampling unit) · **S**ource list (frame) · **S**ize of sample · **P**arameters of interest · **B**udgetary constraint · **P**rocedure (sampling procedure)
Causes of systematic bias	**I** **D**o **N**ot **I**nvite **N**onsense	**I**nappropriate sampling frame · **D**efective measuring device · **N**on-respondents · **I**ndeterminacy principle · **N**atural bias in reporting
Good sample design	**R**ahul **S**hould **V**isit **C**hennai **A**gain	**R**epresentative · **S**mall sampling error · **V**iable within funds · **C**ontrols systematic bias · **A**pplies to the universe with confidence
Basic sampling designs (2 × 2 chart)	**S**weet **C**hai **C**ools **P**eople	**S**imple random (probability, unrestricted) · **C**omplex random (probability, restricted) · **C**onvenience (non-probability, unrestricted) · **P**urposive (non-probability, restricted)
Probability vs non-probability (table rows)	**S**ome **C**ats **S**leep **B**eside **C**ows	**S**election · **C**hance of inclusion · **S**ampling error · **B**ias · **C**ost
Levels of measurement	**N**o **O**ne **I**s **R**ight	**N**ominal · **O**rdinal · **I**nterval · **R**atio
Sources of measurement error	**R**ats **S**teal **M**y **I**dli	**R**espondent · **S**ituation · **M**easurer · **I**nstrument
Important sampling distributions	**M**y **P**arents **T**ook **F**ive **C**hocolates	**M**ean · **P**roportion · **t** (Student's t) · **F** · **C**hi-square
Uses of standard error	**C**ats **R**eally **L**ove	**C**hance: tests if a difference is due to chance · **R**eliability and precision of the sample · **L**imits for the population parameter
Rating scale: merits	**C**ool **L**ions **I**n **W**hite **M**asks	**C**ompares favourably with other methods · **L**ess time · **I**nteresting to use · **W**ide range of applications · **M**any variables
Structured vs unstructured (table rows)	**Q**ueens **R**arely **C**ook **F**ish	**Q**uestions · **R**ecording · **C**omparability · **F**reedom of interviewer

The one-page cheat sheet

Types of research

Dad Always Asks For Quick Questions, Clear Explanations

Descriptive vs Analytical · Applied vs Fundamental · Quantitative vs Qualitative · Conceptual vs Empirical, plus one-time vs longitudinal, field, lab or simulation, clinical, conclusion vs decision-oriented

Ethics: 7 principles

Honest Officers In Court Openly Respect Confidentiality

Honesty · Objectivity · Integrity · Carefulness · Openness · Respect for IP · Confidentiality

Misconduct

Fake Facts Punished Fabrication · Falsification · Plagiarism. Not honest error.

Plagiarism: direct, self, mosaic, paraphrasing, accidental, misattribution

Citation functions

Very Accurate Data Verification · Acknowledgment · Documentation

Failed citations:

Some Bad Scholars Cheat Spurious · Biased · Self · Coercive

Literature review

Existing knowledge gives: context · significance · originality · tools.

Prior art, 5 steps: hypothesis → models and conditions → connections → compare and contrast → strong points and loopholes

Title

3 aspects: **type · length · specific markers**, plus keywords. Title = marketing + traceability. Critical reading finds mistakes (easy); creative reading finds ideas (hard).

Variables

IV = cause (experimental or subject) · DV = effect · Extraneous = other influence · Confounded = can't be separated · Continuous = decimals · Discrete = integers · Control = minimise extraneous

Hypothesis

H_0: $\mu_1 = \mu_2$ (no effect) · H_1: $\mu_1 \neq \mu_2$ · Directional: $\mu_1 < \mu_2$ · Non-directional: ≠

By formulation vs by direction

Design principles

Cats Run Rapidly Control · Randomisation · Replication

Blocking ladder: **CRD 0 → RBD 1 → LSD 2** blocking factors · Factorial = 2+ factors

Models

CRD: Y = μ + T + e

RBD: Y = μ + T + B + E

LSD: Y = μ + T + R + C + E

Factorial: combinations = product of the levels (3 × 2 × 2 = 12)

Slide examples

CRD: baking powder (¼ to 1 tbsp) · RBD: fertilisers × crops · LSD: strawberries (drip, overhead, none; moisture × nitrogen) · Factorial: dosage × duration (2 × 2); therapy × duration × gender (3 × 2 × 2)

Research design

Need: smooth sailing · advance planning · reliability · direction · decisions · no blind search · anticipate problems

Good design: minimum bias, maximum reliability, flexible, controls extraneous variables

Data collection

Primary:

Only Idiots Question Sachin

Observation · Interview · Questionnaire · Schedule

Secondary check:

Rahul Scores Always Reliability · Suitability · Adequacy

Need for sampling

Time and money · accuracy · infinite population · destructive testing · estimate sampling error

Precision: ₹4000 ± 4% = ₹3840 to ₹4160 · 95% confidence = 5% significance

Systematic bias

I Do Not Invite Nonsense

Inappropriate frame · Defective device · Non-respondents · Indeterminacy · Natural bias

A bigger sample fixes sampling error, not bias

Good sample design

Representative · small sampling error · within funds · bias controlled · generalisable with confidence

Steps: universe → unit → frame → size → parameters → budget → procedure

Sample types (2 × 2)

Probability + unrestricted = simple random · probability + restricted = complex random (cluster, systematic, stratified)

Non-probability + unrestricted = convenience · non-probability + restricted = purposive (quota, judgement)

Scales and errors

No One Is Right Nominal (labels) · Ordinal (order) · Interval (equal gaps, no zero) · Ratio (true zero)

Measurement errors:

Rats Steal My Idli Respondent · Situation · Measurer · Instrument

Final reminders

Read all three questions first and pick the two with your strongest 6-markers. Differences go in a table. Designs get a diagram. Definitions use slide wording. Underline keywords. Never leave a part blank. All the best.